

Products of Dilatations

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Products of Dilatations

DRAGOMIR Z. DJOKOVIC

Let V be a right vector space over a division ring F . An invertible linear transformation $A: V \rightarrow V$ is called a *dilatation* if there exists a direct decomposition $V = H \oplus L$ into A -invariant subspaces such that the restriction of A to H is the identity map, the restriction of A to L is not the identity and $\dim L = 1$. This definition is equivalent to the definition of dilatations in [1, p. 3-4]. The subspaces H and L are uniquely determined by A and we shall write $H = H_A$, $L = L_A$ when we wish to indicate this dependence. Note that the inverse of a dilatation is also a dilatation.

When $\dim V = n$ is finite we shall identify $\text{Aut}(V)$ with $GL_n(F)$. In that case an invertible matrix A is a dilatation if it is similar over $GL_n(F)$ to a diagonal matrix whose diagonal entries are $1, 1, \dots, 1, \lambda$ where $\lambda \neq 1$. If $\dim V \geq 1$ and if F has at least 3 elements then $\text{Aut}(V)$ contains dilatations. If F is the field of two elements then there are no dilatations. Our objective is to prove the following

Theorem. *If F has at least four elements then every $A \in GL_n(F)$ is a product of at most n dilatations and this bound is best possible.*

Proof. We shall proceed by induction on n . If $n = 0$ or 1 the assertion is obvious.

Case $n = 2$. If A is diagonalizable the assertion is trivial. Otherwise we may assume that A is in the canonical form

$$A = \begin{pmatrix} 0 & \alpha \\ 1 & \beta \end{pmatrix}.$$

Since F has at least four elements we can choose $\lambda \in F$ distinct from $0, 1, -\alpha^{-1}$. Let

$$\mu = (1 - \beta)\alpha^{-1} - \lambda.$$

We have

$$\begin{pmatrix} \lambda & 0 \\ \mu & 1 \end{pmatrix} A = \begin{pmatrix} \lambda & 0 \\ \mu & 1 \end{pmatrix} \begin{pmatrix} 0 & \alpha \\ 1 & \beta \end{pmatrix} = \begin{pmatrix} 0 & \lambda\alpha \\ 1 & \mu\alpha + \beta \end{pmatrix},$$

and it is easy to check that both

$$B = \begin{pmatrix} \lambda & 0 \\ \mu & 1 \end{pmatrix} \quad \text{and} \quad C = \begin{pmatrix} 0 & \lambda\alpha \\ 1 & \mu\alpha + \beta \end{pmatrix}$$

are dilatations. For C , for instance, this can be proved by observing that

$$C \begin{pmatrix} \lambda\alpha \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda\alpha \\ 1 \end{pmatrix}, \quad C \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} (-\lambda\alpha)$$

and that $-\lambda\alpha \neq 1$.

Case $n \geq 3$. We consider A as a linear operator on $V = F^n$. It suffices to show that there exists a dilatation B such that V admits a direct decomposition $V = W \oplus D$ where W and D are BA -invariant, $\dim D = 1$ and the restriction of BA to D is the identity map. Indeed, if this is so we can apply the induction hypothesis to the restriction of BA to W . It is also clear that we may assume that V is cyclic for A . Therefore we can assume that

$$A = \begin{bmatrix} 0 & & & \alpha_1 \\ 1 & 0 & & \alpha_2 \\ & 1 & \ddots & \vdots \\ & & & 0 \\ & & & 1 & \alpha_n \end{bmatrix}.$$

Let $e_1, \dots, e_n$ be the standard basis of $V = F^n$. Choose $\lambda \in F$ distinct from 0, 1 and put

$$\begin{aligned} \mu &= \lambda + 1 + \alpha_2 \alpha_1^{-1}, \\ a &= (e_1 + e_2(\mu - 1) - e_3\mu)(\lambda - 1)^{-1}, \\ b &= (-e_1 + e_2(\lambda - \mu) + e_3\mu\lambda)(\lambda - 1)^{-1}. \end{aligned}$$

Let B be the dilatation such that $H_B = \langle b, e_3, e_4, \dots, e_n \rangle$, $L_B = aF$, $Ba = a\lambda$. Let $u = a + b = e_2 + e_3\mu$, $v = a\lambda + b = e_1 + e_2\mu$. Then $Bu = v$ and we have

$$BAv = BA(e_1 + e_2\mu) = B(e_2 + e_3\mu) = Bu = v$$

We claim that the subspace $W = \langle e_2, e_3, \dots, e_n \rangle$ is BA -invariant. For this it suffices to verify that $BAe_n \in W$. We have

$$BAe_n = B(e_1\alpha_1 + e_2\alpha_2) + e_3\alpha_3 + \dots + e_n\alpha_n$$

and it remains to verify that $Be_1\alpha_1 + Be_2\alpha_2 \in W$. From

$$B(e_2 + e_3\mu) = Bu = v$$

we have

$$Be_2 = v - e_3\mu = a\lambda + b - e_3\mu.$$

From

$$B(e_1 + e_2\mu) = Bv = B(a\lambda + b) = a\lambda^2 + b$$

we then obtain

$$\begin{aligned} Be_1 &= a\lambda^2 + b - (v - e_3\mu)\mu \\ &= a\lambda(\lambda - \mu) + b(1 - \mu) + e_3\mu^2. \end{aligned}$$

Using the expressions for a and b in terms of e_1, e_2, e_3 we see that the coefficient of e_1 in $Be_1\alpha_1 + Be_2\alpha_2$ is

$$(\lambda - 1)^{-1}\{\lambda(\lambda - \mu)\alpha_1 + (\mu - 1)\alpha_1 + (\lambda - 1)\alpha_2\} = (\lambda + 1 - \mu)\alpha_1 + \alpha_2 = 0.$$

Hence $V = W \oplus D$ where $D = vF$, both W and D are BA -invariant and $BAv = v$.

This completes the proof of the first assertion.

It remains to prove that our bound is the best possible. Let $A \in GL_n(F)$ be chosen so that $Ax = x$ for $x \in F^m$ implies $x = 0$. For instance, one can take $A = I\lambda$ where $\lambda \neq 0, 1$. Assume that $A = B_1B_2 \cdots B_m$ where the B_i 's are dilatations. Let

$$H_i = H_{B_i}, \quad 1 \leq i \leq m$$

be the hyperplanes associated with these dilatations and let $H = H_1 \cap \cdots \cap H_m$. If $m < n$ then $\dim H \geq 1$ and for every $x \in H$ we have $Ax = x$. This is a contradiction. Therefore we must have $m \geq n$.

Remarks. 1) If F is the Galois field of 3 elements then the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

cannot be written as a product of two dilatations. Indeed, $GL_2(F)$ has 12 dilatations which are listed below

$$\begin{aligned} &\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \\ &\begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix}, \quad \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}, \\ &\begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}. \end{aligned}$$

One can now easily verify our claim.

2) The problem of factorizing a matrix $A \in GL_n(F)$ (F a field) into a product

$$A = B_1B_2 \cdots B_m$$

such that every B_i is a dilatation with determinant -1 (called also reflections or simple involutions) has been considered by H. Radjavi [2]. The condition $\det A = \pm 1$ is necessary and sufficient for the existence of such factorizations

if $\text{char } F \neq 2$. (When $\text{char } F = 2$ Radjavi defines simple involutions in a different way.) He showed that one can always take $m \leq 2n - 1$ but it is not known if this bound is the best possible.

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